

Web Engineering & Development (SWE 363)

Interactive Front-End Development

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Vite + React & JSX — Quick Guide

This page gives you a fast start with **React using Vite**, explains **JSX**, and compares **Vite vs Create React App (CRA)**.

Start a React project with Vite

```
# create a project
npm create vite@latest my-app

# go inside project
cd my-app

# install dependencies
npm install

# run dev server
npm run dev
```

The dev server starts in milliseconds. Open the printed local URL in your browser.

■ What is JSX?

JSX stands for **JavaScript XML**.

- It's a **syntax extension** for JavaScript used in **React**.
- It allows you to write **HTML-like code** inside JavaScript.
- React (via Babel / Vite) **compiles JSX to plain JavaScript**.

Why do we use JSX?

1. **Looks like HTML** → easy to describe UI.
2. **Powerful** → embed JavaScript expressions inside `{ }`.
3. **Faster development** → more readable and maintainable.
4. **Closer to the UI** → build components in a single file.

Examples

Without JSX (pure JavaScript)

```
import React from "react";

function Hello() {
  return React.createElement("h1", null, "Hello, World!");
}

export default Hello;
```

With JSX (cleaner)

```
import React from "react";

function Hello() {
  return <h1>Hello, World!</h1>;
}
export default Hello;
```

Both snippets produce the same UI; JSX is easier to read and write.

Embedding JS in JSX

```
const name = "Hasan";
const age = 25;

function Student() {
  return (
    <div>
      <h2>{name}</h2>
      <p>Age: {age}</p>
    </div>
  );
}
```

Here, `{name}` and `{age}` are **JavaScript expressions** inside JSX.

Vite vs CRA

Feature	Vite	Create React App (CRA)
Speed (Dev server start)	⚡ Very fast (milliseconds, thanks to ES Modules)	🐌 Slow (can take 20–30s for large apps)
Hot Reload (HMR)	Instant updates, keeps component state	Slower reloads, sometimes refreshes the whole page
Project size	Lightweight, fewer dependencies	Heavier, installs many dependencies
Production Build	Smaller, optimized with tree-shaking	Larger, less optimized by default
Ease of Setup	Simple, modern setup with <code>npm create vite</code>	Easy, but older approach with <code>npx create-react-app</code>

✓ Quick Tips

- Use `npm run dev` for development, `npm run build` for production, and `npm run preview` to test the build locally.
- Add Tailwind, ESLint, and Prettier easily via Vite plugins or templates.

Suggested Repo Structure

```
my-app/  
├── src/  
│   ├── assets/  
│   ├── components/  
│   │   └── StudentCard.jsx  
│   ├── App.jsx  
│   └── main.jsx  
├── index.html  
├── package.json  
├── README.md ← (this file)  
└── vite.config.js
```

👁 Example StudentCard.jsx

```
function StudentCard({ name, id, department }) {  
  return (  
    <div style={{  
      border: "1px solid #ccc",  
      borderRadius: "8px",  
      padding: "16px",  
      width: "250px",  
      margin: "10px auto",  
      textAlign: "center",  
      boxShadow: "2px 2px 6px rgba(0,0,0,0.1)"  
    }}>  
      <h3>Name: {name}</h3>  
      <p>ID: {id}</p>  
      <p>Department: {department}</p>  
    </div>  
  );  
}  
  
export default StudentCard;
```

Notes

- JSX requires a **single parent element** to be returned from a component.
- Use `className` instead of `class` in JSX.
- Expressions inside JSX go in `{ }`.

